

MUHAMMAD TAIMOOR TARIQ

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Education

University of Illinois at Urbana-Champaign

PhD. in Computer Science

- Advised by: Radhika Mittal

August. 2022 – Present

IL, USA

Lahore University of Management Sciences (LUMS)

Bachelor of Science in Computer Science

- Placed on the Dean's Honors List for academic excellence for years 2020-2022

Sept. 2018 – May 2022

Lahore, Pakistan

Publications

Enabling Emerging Edge Applications Through a 5G Control Plane Intervention

Mukhtiar Ahmad, Ali Nawazish, M Taimoor Tariq, Basit Iqbal, Taqi Raza, Zafar Ayyub Qazi

ACM CoNEXT 2022, Rome, Italy

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane

Mukhtiar Ahmad, Ali Nawazish, M Taimoor Tariq, Usman Jafri, Adnan Abbass, Mashal Abbas, Zartash Uzmi, Zafar Qazi

IEEE/ACM Transactions on Networking, 2022

Fast Failure Detectors for 5G Edge Deployments

M Taimoor Tariq, Maha Kamal, Vafa Batool

Student Research Competition, SOSP 2021

Research Experience

Slice-Aware Interference Management For 5G Networks

Research Assistant - Advisor: Dr. Radhika Mittal

- Worked on projects related to Slice-aware Interference Management and Load Balancing for Multi Cell Deployments.
- Under submission at Mobicom 2025

August-2022 - Present

Urbana, IL

Fast Failure Detectors for 5G Edge Deployments

Research Assistant - Advisor: Dr. Zafar Ayyub Qazi

- Goal was to design an efficient system to detect failures in the cellular control plane
- Responsible for leading the design and implementation of a system based on distributed heartbeats
- Results showed a reduction in initial detection time by 50% and confirmation time by 30%, compared to existing systems

March 2021 – May 2022

Lahore, Pakistan

Enabling Edge Applications Through a 5G Control Plane Intervention

Research Assistant - Advisor: Dr. Zafar Ayyub Qazi

- Goal was to design a fault tolerant control plane by deploying replicated control plane nodes at the edge
- Implemented and evaluated the system against prior works (DPCM, ECHO, SkyCore) under varying network conditions
- Results showed an improvement of up to 3.8x in Procedure Completion Times under failure conditions

June 2021 – September 2021

Lahore, Pakistan

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane

Research Assistant - Advisor: Dr. Zafar Ayyub Qazi

- Goal was to minimize latency in cases of user mobility by implementing a geo-aware, proactive state replication system
- Implemented a consistent-hashing based replication scheme which ensures the read-your-writes consistency model
- Results showed a reduction in Procedure Completion Times by up to 3.1x in cases of user mobility

June 2020 – March 2021

Lahore, Pakistan

Projects

Apnay Rung | ReactJS, NodeJS

- Built a UX-friendly React based frontend catering to diverse user roles e.g.Admin, Sellers, Customers
- Built and deployed a PostgreSQL Database for backend
- Built a REST API for efficient backend-frontend communication

March 2021 – April 2021

Raft-based Key Value Store | Go

- Built a fault-tolerant distributed data storage system based on Raft

October 2020 – November 2020

Teaching Experience

Teaching Assistant

CS-425: Distributed Systems

August 2024 – Present

UIUC, IL

Teaching Assistant

CS-678: Topics in Internet Research (Graduate)

January 2022 – May 2022

LUMS, Pakistan

Lead Teaching Assistant

CS-582: Distributed Systems (Graduate)

September 2021 – December 2021

LUMS, Pakistan

Teaching Assistant

CS-382: Network-Centric Computing (Undergraduate)

January 2021 – May 2021

LUMS, Pakistan

Technical Skills

Languages: C, C++, Python, Go

Frameworks/Tools: ZooKeeper, DPDK, Wireshark, NodeJS, ReactJS